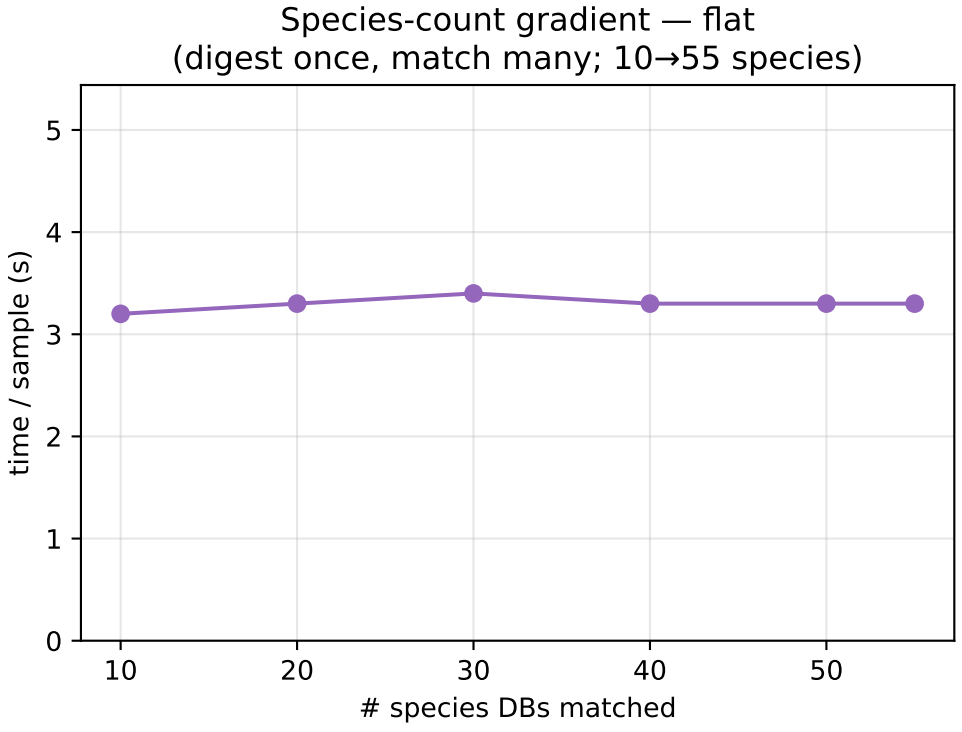
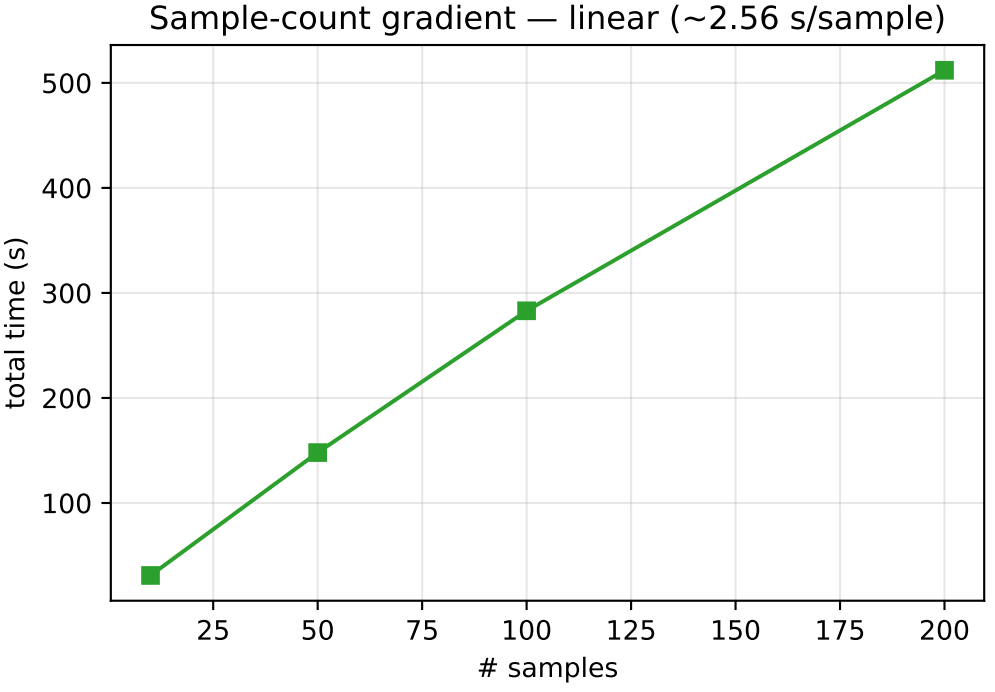
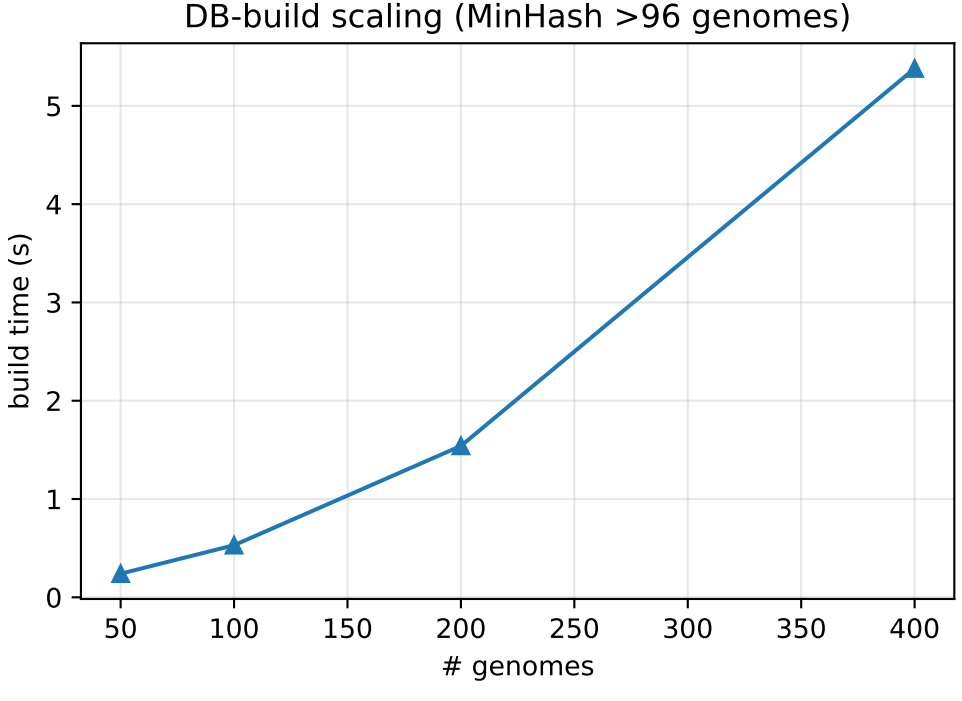
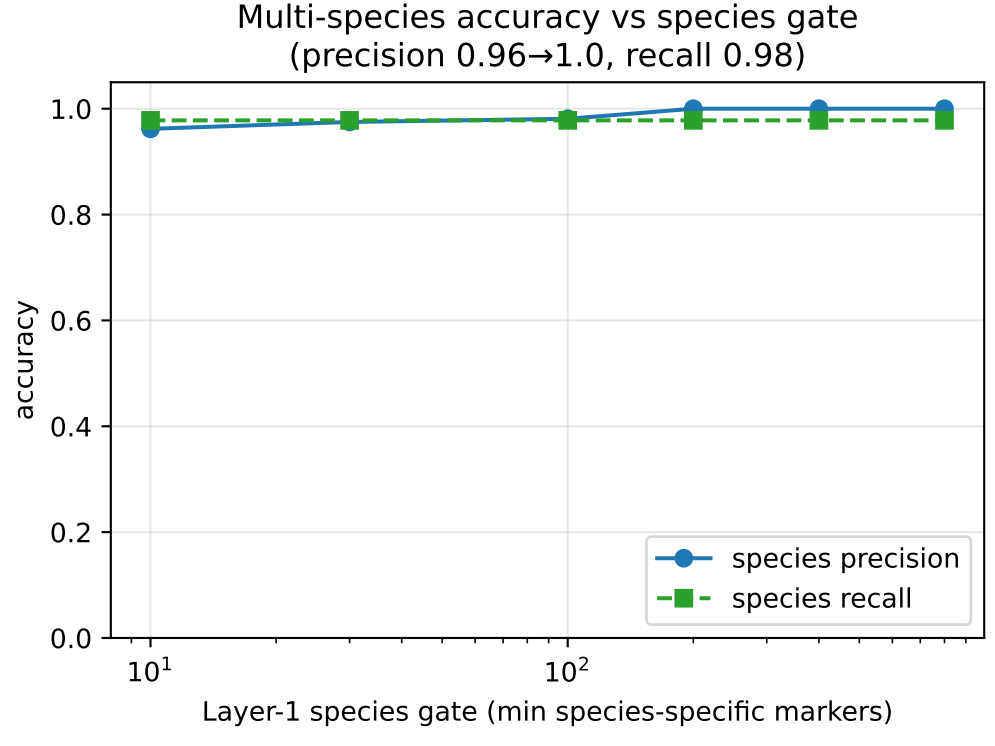

Species-count gradient — flat
(digest once, match many; 10→55 species)
time / sample (s)
species DBs matched
0
1
2
3
4
5
10
20
30
40
50


Sample-count gradient — linear (~2.56 s/sample)
total time (s)
samples
100
200
300
400
500
25
50
75
100
125
150
175
200


DB-build scaling (MinHash >96 genomes)
build time (s)
genomes
0
1
2
3
4
5
50
100
150
200
250
300
350
400


Multi-species accuracy vs species gate
(precision 0.96→1.0, recall 0.98)
accuracy
Layer-1 species gate (min species-specific markers)
species precision
species recall
0.0
0.2
0.4
0.6
0.8
1.0
10^1
10^2